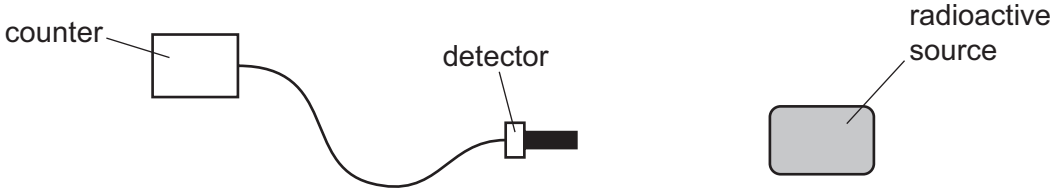
37 An experiment is done to measure the radiation from a radioactive source. The source has a half-life of 10 minutes.

The source is placed close to a detector that is connected to a counter, as shown.



The average background count rate is 20 counts / minute.

At the start of the experiment, the count rate recorded by the counter is 1000 counts / minute.

What is the count rate 10 minutes later?

A 490 counts / minute

B 500 counts / minute

C 510 counts / minute

D 530 counts / minute